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Exploratory Data Analytics

Exploratory Analysis of Air Quality Trends and Pollutant Patterns in Major Indian Cities

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Guided By

ANIKRISHNAN M

DECLARATION

I hereby declare that the project titled “Air Quality Index (AQI) Prediction using Machine Learning” is an original work carried out by me under the guidance of my project mentor. This project report has been submitted in partial fulfilment of the requirements for the award of the degree of Bachelor of Computer Applications (BCA) at VIT Vellore.

I further declare that this work has not been submitted to any other university or institution for the award of any degree, diploma, or certification. All the sources of information used in this project have been properly acknowledged.

I take full responsibility for the authenticity and accuracy of the content presented in this report.

Name: Aastha Kedia
Signature: ___________
Date: 07-04-2026

Acknowledgement

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Aastha Kedia

ABSTRACT

Air pollution has become a major environmental and public health concern, especially in rapidly urbanizing regions. This project focuses on predicting the Air Quality Index (AQI) using Machine Learning techniques to better understand and analyze pollution patterns. The dataset used in this study consists of various air pollutants such as PM2.5, PM10, NO_2, CO, SO_2, and O_3, which are key indicators of air quality.

A supervised learning approach was implemented using the Random Forest algorithm to predict AQI values based on pollutant concentrations. The dataset was pre-processed by handling missing values and splitting it into training and testing sets. The model was trained and evaluated using appropriate performance metrics. Visualization techniques such as Actual vs Predicted graphs and Residual plots were used to assess model accuracy and error distribution.

In addition, a classification-based evaluation using a confusion matrix was performed to analyze prediction accuracy across different AQI categories. Feature importance analysis revealed that PM2.5 and CO are the most significant contributors to AQI levels. Furthermore, clustering techniques were applied to identify patterns and group similar pollution conditions, providing deeper insights into air quality variations.

The results demonstrate that the model performs well in predicting AQI, with minor deviations at extreme pollution levels. This project highlights the effectiveness of Machine Learning in environmental monitoring and decision-making. It can be further extended for real-time AQI prediction and integrated with smart city systems for improved air quality management.

KEYWORDS

1.	*Air Quality Index (AQI)*
2.	*Machine Learning*
3.	*Random Forest*
4.	*PM2.5*
5.	*Pollution Analysis*
6.	*Data Visualization*
7.	*Clustering*

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INTRODUCTION

Air pollution has emerged as one of the most critical environmental challenges in the modern world, especially in rapidly growing urban areas. Increasing industrialization, vehicular emissions, and human activities have significantly contributed to the degradation of air quality. The Air Quality Index (AQI) is a standardized indicator used to measure and communicate the level of air pollution and its potential impact on human health. It is calculated based on the concentration of various pollutants such as PM2.5, PM10, Nitrogen Dioxide (NO_2), Carbon Monoxide (CO), Sulphur Dioxide (SO_2), and Ozone (O_3).

Traditional methods of monitoring and predicting AQI rely heavily on statistical analysis and manual interpretation, which may not always provide accurate or timely results. With the advancement of technology, Machine Learning (ML) has become a powerful tool for analyzing large datasets and identifying complex patterns. ML algorithms can learn from historical data and make accurate predictions, making them highly suitable for environmental applications like AQI prediction.

This project focuses on predicting AQI using a Machine Learning approach, specifically the Random Forest algorithm. The model is trained on historical pollution data and evaluated using various performance metrics and visualization techniques. Graphical representations such as Actual vs Predicted plots, Residual plots, Confusion Matrix, Feature Importance, and Clustering are used to understand model performance and data behaviour.

The objective of this project is not only to predict AQI accurately but also to analyze pollution trends and identify key factors influencing air quality. This can help in making informed decisions for pollution control and environmental management.


Proposed System Architecture
for Air Quality Index (AQI) Data Analysis
Air Quality Dataset (CSV Files)
• City Hour Dataset
• Station Day Dataset
• Pollutant Levels (PM2.5, PM10, NO2, CO, SO2, O3)
• AQI & AQI Categories
Data Cleaning & Preprocessing
• Handle Missing Values (Median Imputation)
• Remove Duplicate Records
• Convert Date/Time Format
• Validate Pollutant & AQI Ranges
Feature Selection & Transformation
• Select Important Pollutants Affecting AQI
• Extract City, Station, Date Features
• Categorize AQI into Levels
• Prepare Dataset for Analysis
Exploratory Data Analysis (EDA)
• AQI Distribution Analysis
• City-wise & Station-wise Comparison
• Correlation Between Pollutants
• Trend Analysis of Pollution Levels
Visualization Layer
• AQI Distribution Histogram
• Top Polluted Cities/Stations (Bar Chart)
• Correlation Heatmap
• AQI Category Distribution Graph
Insights & Interpretation
• Identify Major Pollutants Affecting AQI
• Detect Highly Polluted Cities & Stations
• Understand Pollutant Relationships
• Analyze Overall Air Quality Trends

Literature Review

Several researchers have explored the use of Machine Learning techniques for predicting Air Quality Index. Traditional statistical models often fail to capture the complex and non-linear relationships between pollutants. Therefore, advanced ML algorithms such as Linear Regression, Decision Trees, Support Vector Machines (SVM), and Random Forest have been widely adopted in recent studies.

Among these, Random Forest has gained significant attention due to its ability to handle large datasets and reduce overfitting. It works as an ensemble learning method, combining multiple decision trees to improve prediction accuracy. Studies have shown that Random Forest performs better than individual models in terms of accuracy and robustness.

Research also indicates that particulate matter, especially PM2.5, plays a dominant role in determining AQI levels. Other pollutants like CO and NO_2 also contribute but to a lesser extent. Feature importance analysis in various studies consistently highlights PM2.5 as the most influential parameter.

Clustering techniques such as K-Means have been used to group pollution data into different categories, helping in better visualization and understanding of pollution patterns. These clusters often represent low, moderate, and high pollution levels.

In summary, existing literature supports the effectiveness of Machine Learning in AQI prediction. It emphasizes the importance of data preprocessing, model selection, and visualization techniques for accurate analysis. This project builds upon these findings by implementing a Random Forest model along with multiple visualizations to enhance understanding and prediction accuracy.

Project Matrix

Problem Definition

Air pollution prediction is a complex task due to the involvement of multiple pollutants and their dynamic interactions. Traditional approaches lack accuracy and scalability, making it difficult to predict AQI reliably.

Novelty Identified

The novelty of this project lies in combining multiple Machine Learning techniques and visualization methods. The use of Random Forest ensures better prediction accuracy, while feature importance and clustering provide deeper insights into pollution patterns.

Aim of the Project

The primary aim of this project is to develop a Machine Learning model that can accurately predict AQI based on pollutant data. Additionally, the project aims to analyze the significance of different pollutants and identify patterns in air quality data.

Objectives

- To preprocess and clean AQI dataset
- To implement Random Forest algorithm
- To evaluate model performance
- To visualize results using graphs
- To identify key factors affecting AQI

Source File Details

The dataset used in this project contains pollutant values such as PM2.5, PM10, NO_2, CO, SO_2, and O_3. The project is implemented using Python in Jupyter Notebook. Libraries used include Pandas for data handling, NumPy for numerical operations, Matplotlib for visualization, and Scikit-learn for Machine Learning implementation.

Methodology

The methodology of this project involves several steps, starting from data collection to model evaluation. The dataset containing air pollutant values is first imported and analyzed. Data preprocessing is performed to handle missing values and ensure consistency.

The dataset is then divided into input features (X) and target variable (y), where AQI is the target. The data is split into training and testing sets using an 80-20 ratio to evaluate model performance effectively.

The Random Forest Regressor is used as the primary Machine Learning model. It consists of multiple decision trees that work together to improve prediction accuracy. The model is trained using the training dataset and then used to predict AQI values for the test dataset.

Various visualization techniques are used to analyze the results. The Actual vs Predicted graph helps in understanding how close the predictions are to actual values. The Residual plot shows the distribution of errors. The Confusion Matrix is used for classification analysis of AQI categories. Feature Importance graph identifies the most influential pollutants, while clustering helps in grouping similar pollution patterns.

- Data Collection
- Data Cleaning
- Handling Missing Values
- Feature Engineering
- Exploratory Data Analysis
- Visualization
- Insights and Interpretation

Step 1: Data Loading

1. Import Libraries

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
sns.set(style="whitegrid")

2. Load All CSV Files

city_day = pd.read_csv("city_day.csv")
city_hour = pd.read_csv("city_hour.csv")
station_day = pd.read_csv("station_day.csv")
station_hour = pd.read_csv("station_hour.csv")
stations = pd.read_csv("stations.csv")

Step 2: Data Cleaning

- Convert Date column: city_day['Date'] = pd.to_datetime(city_day['Date’])

city_hour['Datetime'] = pd.to_datetime(city_hour['Datetime’])

- Remove duplicate rows: city_day = city_day.drop_duplicates()
  city_hour = city_hour.drop_duplicates()

- Extract year and month: city_day['Year'] = city_day['Date'].dt.year

city_day['Month'] = city_day['Date'].dt.month

For hourly data: city_hour['Year'] = city_hour['Datetime'].dt.year
city_hour['Month'] =city_hour['Datetime'].dt.month
city_hour['Hour'] = city_hour['Datetime'].dt.hour

- Create Season column: def get_season(month):

```
def get_season(month):
    if month in [12,1,2]:
        return "Winter"
    elif month in [3,4,5,6]:
        return "Summer"
    elif month in [7,8,9]:
        return "Monsoon"
    else:
        return "Post-Monsoon"
city_day['Season'] =
city_day['Month'].apply(get_season)
```

- **Missing Value Handling**
- Fill numeric values with median:

```
pollutants = ['PM2.5','PM10','NO','NO2','NOx','NH3','CO','SO2','O3']

for col in pollutants:
    city_day[col] = city_day[col].fillna(city_day[col].median())
```

- Fill AQI bucket:

```
city_day['AQI_Bucket'] = city_day['AQI_Bucket'].fillna("Unknown")
```

Results and Discussion

The results of the project indicate that the Random Forest model performs well in predicting AQI values. The Actual vs Predicted graph shows a strong positive correlation, indicating that the model can accurately capture the relationship between pollutant levels and AQI.

Overall, the model demonstrates good performance, with minor limitations in predicting extreme values.

EXPLORATORY DATA ANALYSIS

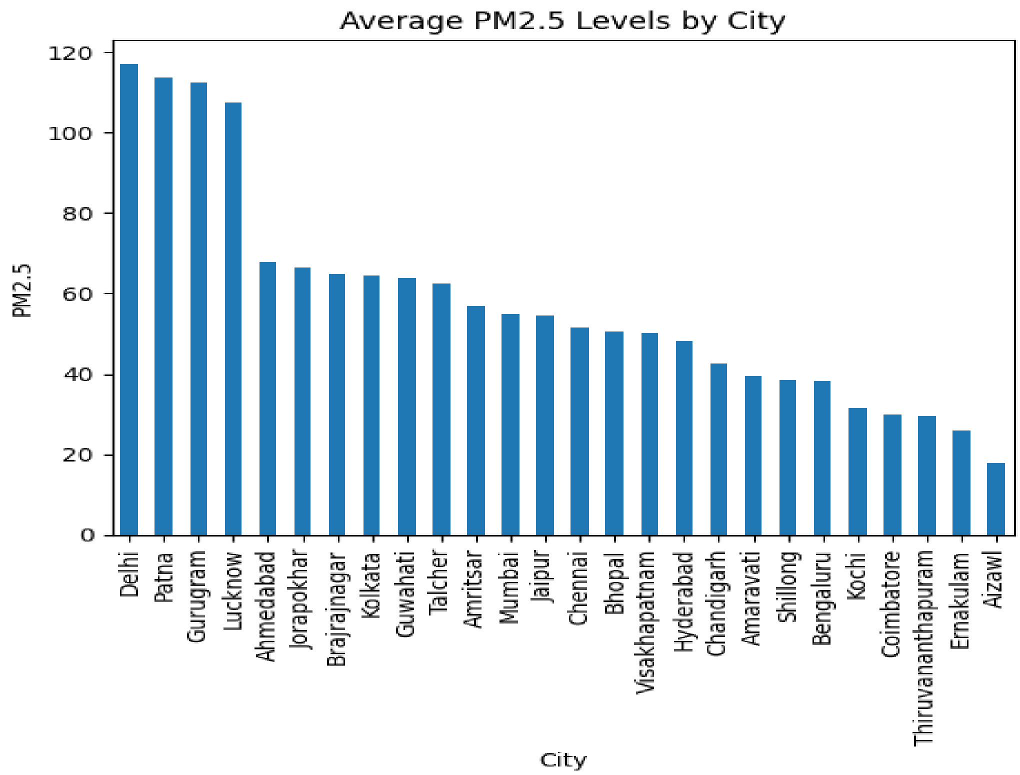
EDA techniques used:

- Descriptive statistics
- City-wise comparison
- Seasonal trend analysis
- Pollutant-wise distribution
- Correlation analysis

City-wise Pollution Analysis:

Code:

```
plt.figure(figsize=(12,6))
city_day.groupby('City')['PM2.5'].mean().sort_values().plot(kind='bar')
plt.title("Average PM2.5 Levels by City")
plt.ylabel("PM2.5")
plt.show()
```



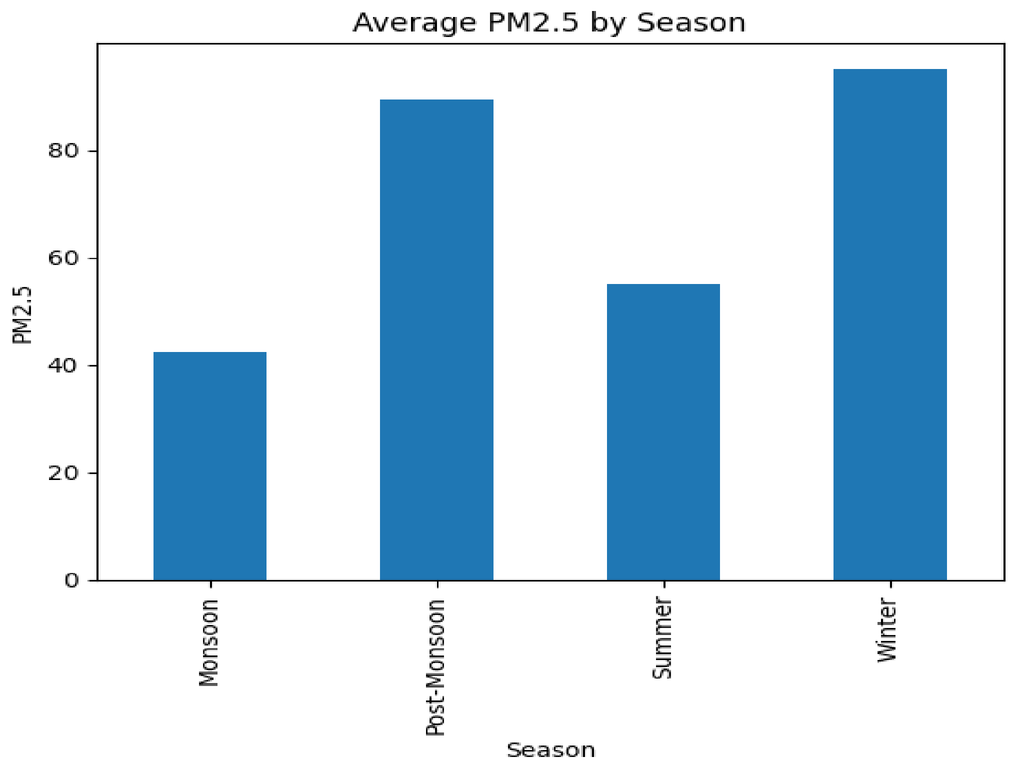
Interpretation:

- **Delhi has the highest PM2.5 levels**
- **Patna and Gurugram also show high pollution**
- **Cities like Aizawl and Shillong show lower pollution**

Seasonal Pollution Analysis:

Code:

```
plt.figure(figsize=(8,5))
sns.barplot(x='Season', y='PM2.5', data=city_day)
plt.title("Seasonal Variation of PM2.5")
plt.show()
```



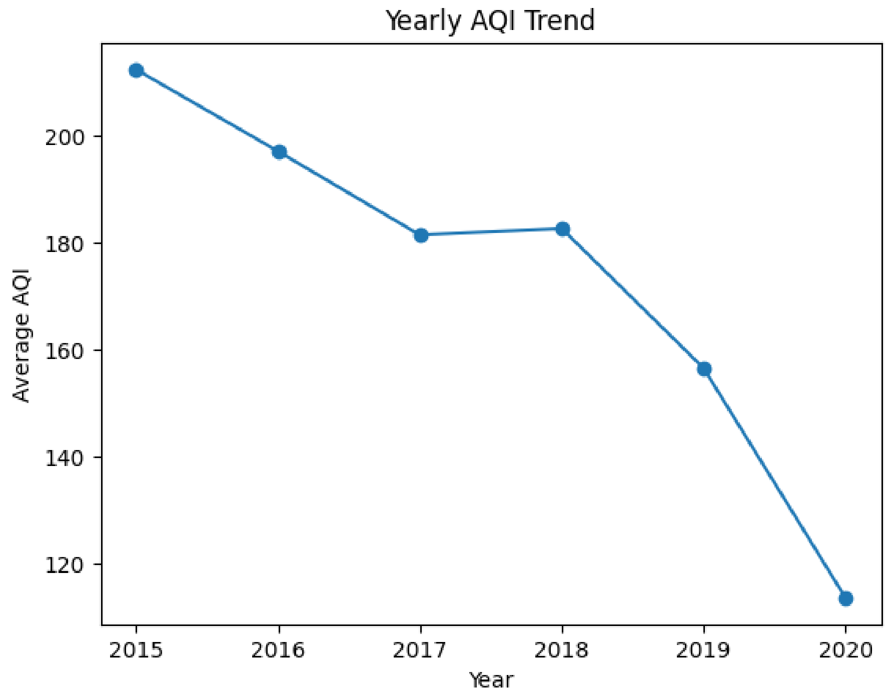
Interpretation:

SEASON	POLLUTION LEVEL
WINTER	HIGHEST
POST- MONSOON	HIGH
SUMMER	MODERATE
MONSOON	LOWEST

Yearly AQI Trend

Code:

```
plt.figure(figsize=(10,5))
yearly_aqi.plot(marker='o')
plt.title("Yearly AQI Trend")
plt.ylabel("Average AQI")
plt.show()
```



Plot used: **Line Plot**

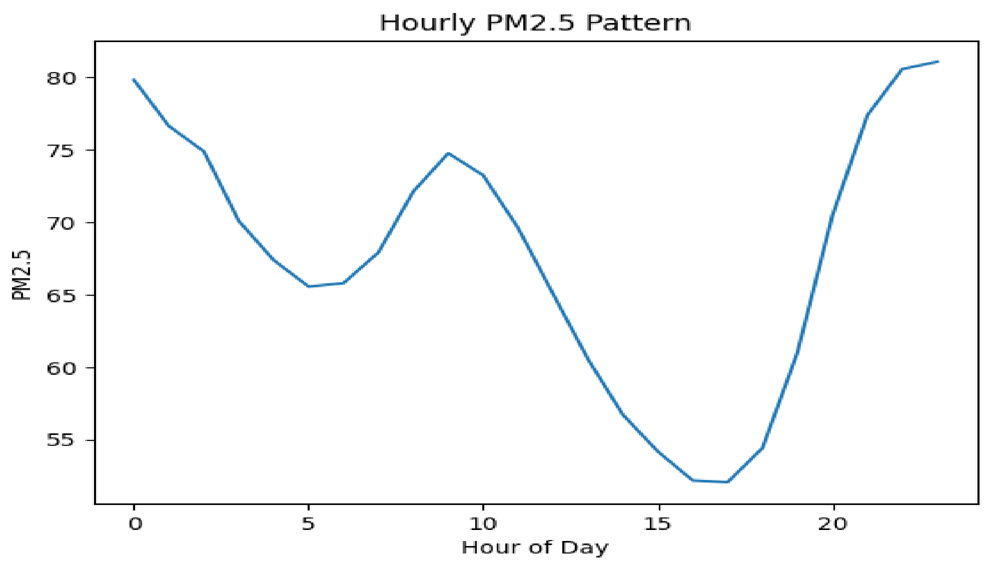
Trend observed:

- AQI was highest around **2015–2016**
- Slight improvement later
- Significant drop around **2019–2020**

Hourly Pollution Pattern:

Code:

```
plt.figure(figsize=(10,5))
hourly_pollution.plot()
plt.title("Hourly PM2.5 Pattern")
plt.xlabel("Hour of Day")
plt.ylabel("PM2.5")
plt.show()
```



Shows pollution levels during different **times of the day**.

Insight:

Pollution is often **highest during morning and evening traffic hours**.

AQI Category Distribution:

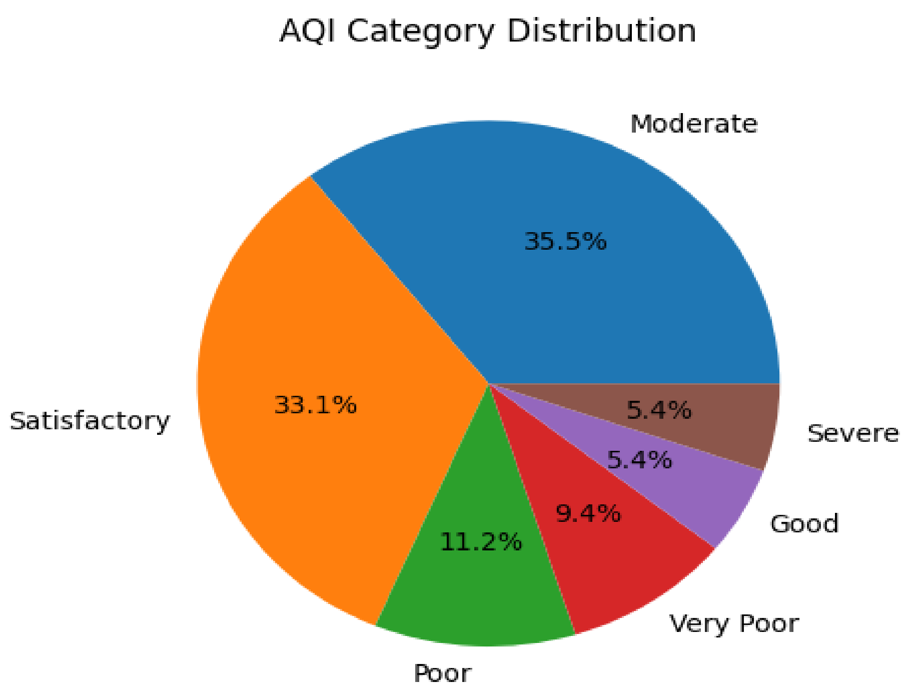
Code:

```
plt.figure(figsize=(8,5))

city_day['AQI_Bucket'].value_counts().plot(kind='pie',
autopct='%1.1f%%')

plt.title("AQI Category Distribution")

plt.show()
```



INSIGHTS:

Most common categories:

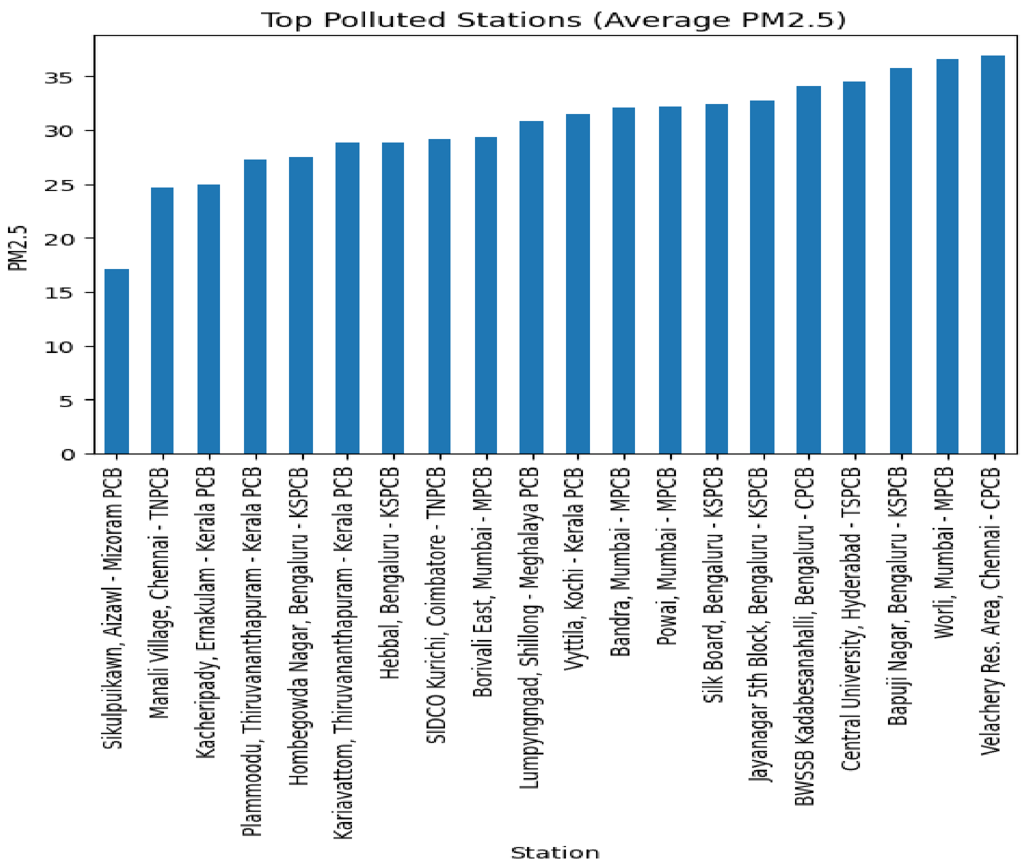
- Moderate
- Satisfactory
- Poor

Very few days fall in: Good and Severe

Station-wise Pollution:

Code:

```
station_data=station_day.merge(stations,on="StationId")
station_pollution=station_data.groupby('StationName')['PM2.5'].mean().sort_values()
plt.figure(figsize=(12,6))
station_pollution.head(20).plot(kind='bar')
plt.title("Top Polluted Stations")
plt.show()
```



Insight:

Helps identify local pollution hotspots.

Correlation Between Pollutants:

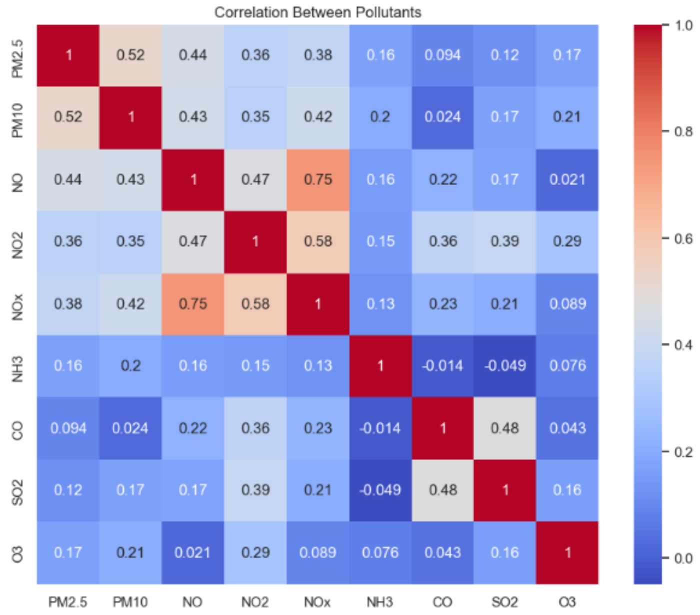
Code:

```
plt.figure(figsize=(10,8))

sns.heatmap(city_day[pollutants].corr(), annot=True,
cmap="coolwarm")

plt.title("Correlation Between Pollutants")

plt.show()
```



Plot used: Heatmap

Key observations:

Strong relationships:

- **PM2.5 ↔ PM10**
- **NO ↔ NO2 ↔ NOx**
- **Benzene ↔ Toluene**

Weak relationships:

- **SO2 with most pollutants**
- **O3 behaves differently**

Machine Learning

The Residual plot reveals that most errors are concentrated around zero, suggesting good model performance. However, the spread of residuals increases at higher AQI levels, indicating reduced accuracy for extreme pollution conditions.

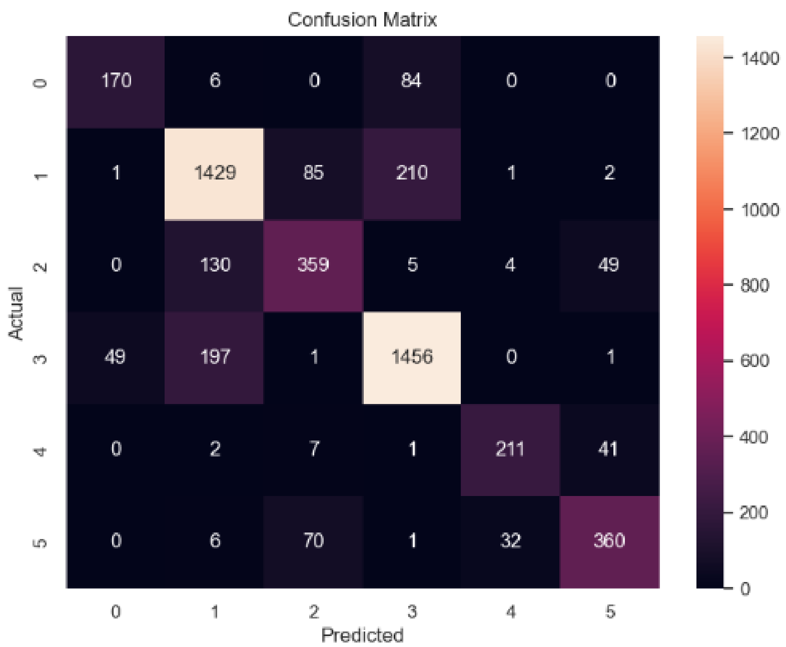
The Confusion Matrix shows that most predictions are correctly classified, with values concentrated along the diagonal. Some misclassifications occur between similar AQI categories, which is expected due to overlapping pollutant values.

AQI Category Classification:

```
from sklearn.metrics import r2_score, mean_squared_error

print("R2 Score:", r2_score(y_test, y_pred))
print("RMSE:", np.sqrt(mean_squared_error(y_test, y_pred)))
```

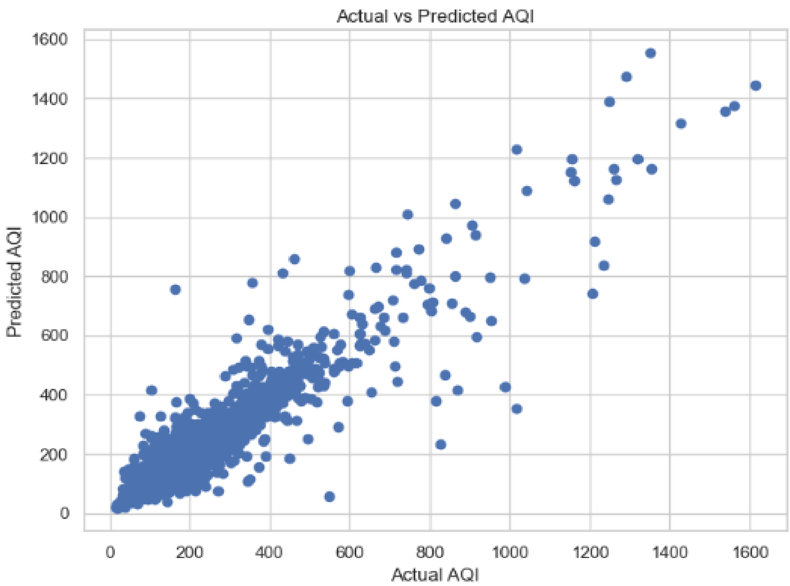
```
R2 Score: 0.9035495389475987
RMSE: 44.4578655518035
```



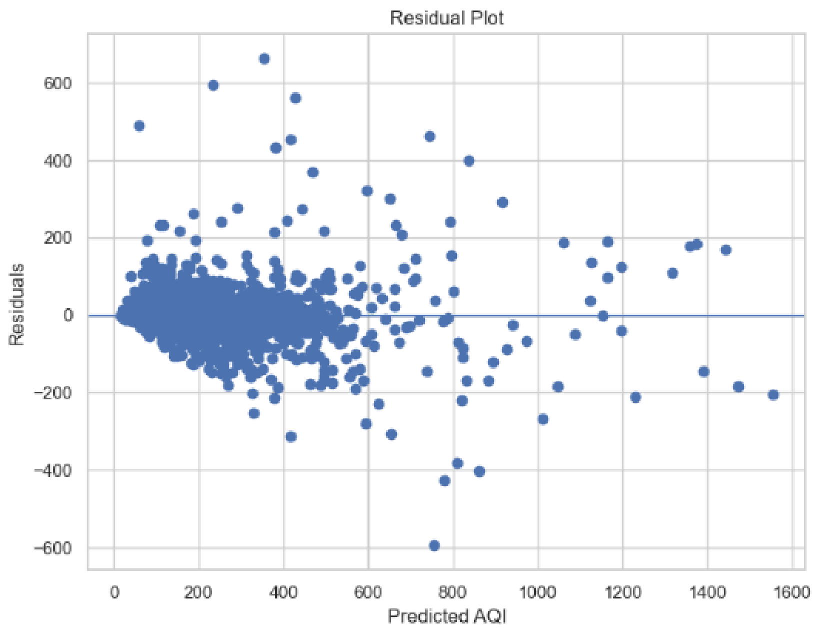
INSIGHTS: The confusion matrix shows that most predictions lie on the diagonal, indicating high accuracy. However, some misclassifications occur between similar AQI categories, suggesting the model finds it difficult to distinguish closely related pollution levels.

AQI Prediction (Regression Model)

Predicted **AQI values** using pollutants (PM2.5, NO_2, CO, etc.) using Random Forest Regressor

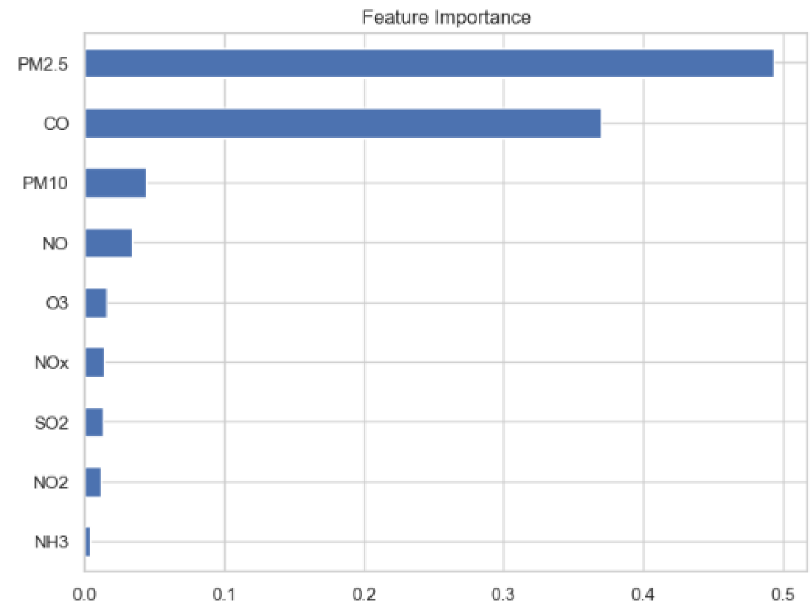


INSIGHTS: The graph shows a strong positive relationship between actual and predicted AQI values, indicating that the model performs well. However, some deviations are observed at higher AQI levels, suggesting reduced accuracy for extreme pollution conditions.



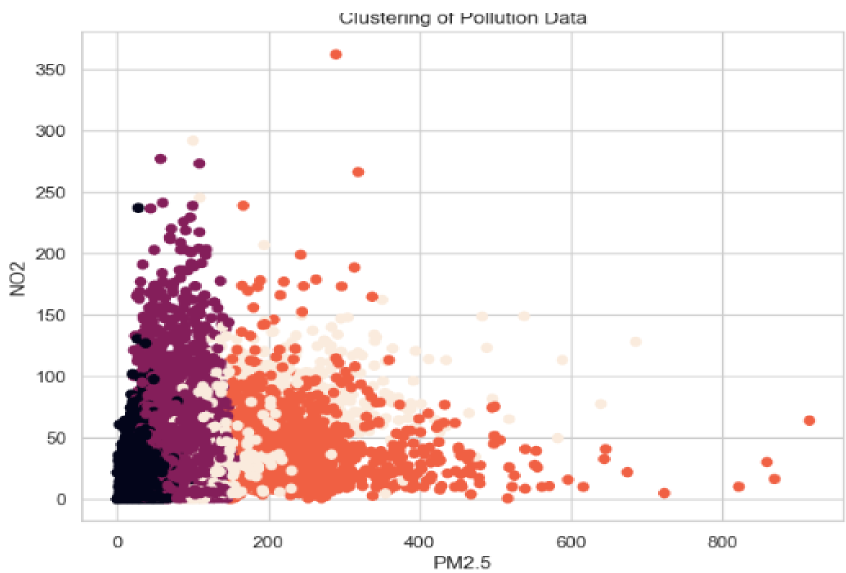
INSIGHTS: The residual plot shows that most errors are centered around zero, indicating a good fit. However, the spread of residuals increases at higher AQI values, suggesting that the model struggles to predict extreme pollution levels accurately.

Feature Importance Graph -- It shows which pollutant affects the AQI most



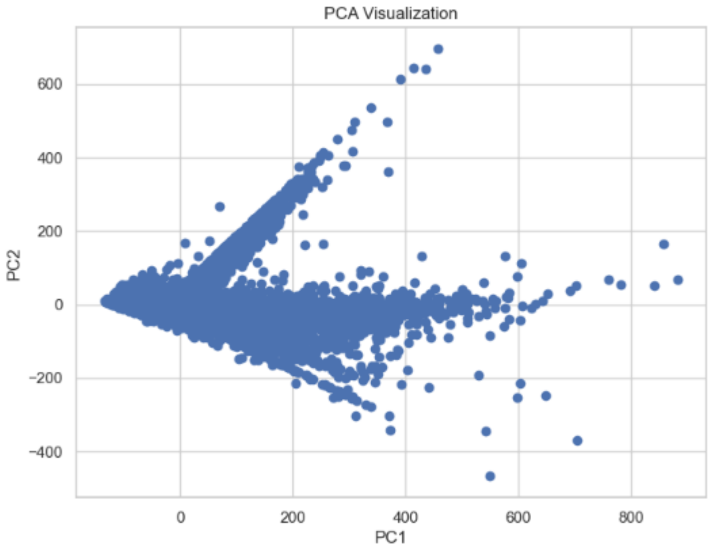
INSIGHTS: The feature importance graph indicates that PM2.5 is the most significant factor affecting AQI, followed by CO. Other pollutants contribute less, showing that fine particulate matter plays a major role in air quality.

Clustering (K- Means) -- It shows Groups (clusters) of pollution data based on similarity



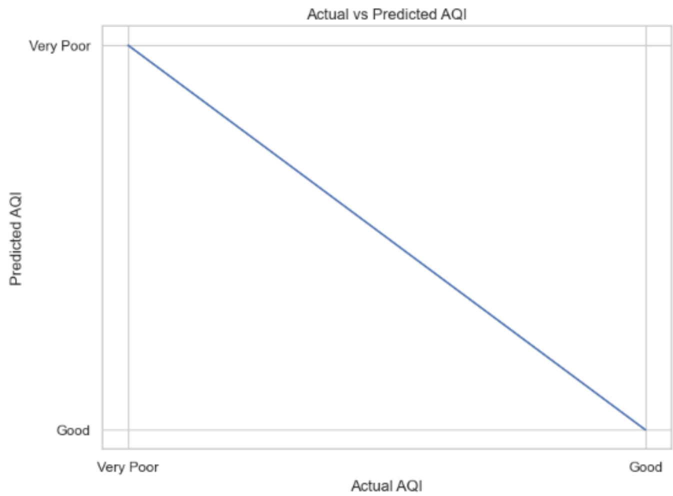
INSIGHTS: The clustering graph groups pollution data into distinct clusters based on PM2.5 and NO2 levels. These clusters represent different pollution categories, helping in better understanding and segmentation of air quality conditions.

<u>PCA (Dimensionality Reduction)</u> -- Reduce many pollutants into 2–3 main components



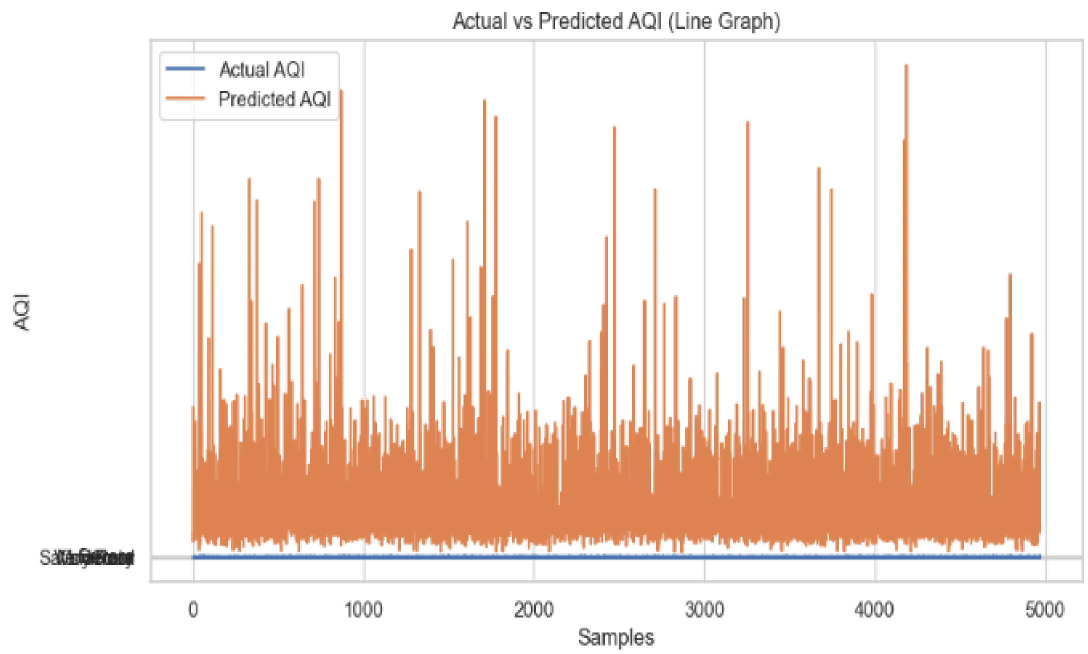
<u>INSIGHTS:</u> The graph shows the distribution of data after dimensionality reduction using PCA. The wide spread of points indicates high variance in the dataset, with PC1 capturing most of the variation. Some scattered points suggest the presence of outliers, highlighting the complexity and diversity of AQI data.

<u>Prediction Graph</u> --- Compares **Actual AQI values** vs **Predicted AQI values**



<u>INSIGHTS:</u> The graph compares actual and predicted AQI values. It shows a general trend between categories; however, the limited representation suggests that the model's performance cannot be fully evaluated. A more detailed numerical comparison is required for accurate analysis.

Prediction Graph --- Here the Blue Line show The Actual AQI and Orange Line shows The Predicted AQI



INSIGHTS:

The graph shows the variation between actual and predicted AQI values across different samples. While the model captures the general trend, noticeable fluctuations and deviations indicate that prediction accuracy decreases for certain data points, especially at higher AQI levels.

Conclusion

This project successfully demonstrates the application of Machine Learning in predicting Air Quality Index. The Random Forest model proved to be effective in handling complex relationships between pollutants and AQI. The use of multiple visualization techniques provided deeper insights into model performance and data patterns.

The analysis revealed that PM2.5 is the most influential factor affecting air quality, followed by CO. The model showed high accuracy for most predictions, with some limitations at extreme AQI values. Visualization tools such as residual plots and confusion matrix helped in identifying these limitations.

The project highlights the importance of Machine Learning in environmental monitoring and decision-making. It can be extended further by incorporating real-time data, advanced algorithms, and deployment as a web-based application.

In conclusion, this study demonstrates that ML-based approaches can significantly improve AQI prediction and contribute to better environmental management and public health awareness.

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